MILITARY SPECIFICATION

NAVIGATION DISPLAY SET, AN/ASN-124

This amendment forms a part of Military Specification MIL-N-85005A(AS), dated 31 May 1978.

PAGE 1

Delete the note on the bottom center of the first page and replace with the following:

"

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

"

PAGE 3

2.1 Delete paragraph 2.1 in its entirety, including the types of publications, and replace it with the following:

"2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-W-5088	Wiring, Aerospace Vehicle
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-T-5422	Testing, Environmental, Airborne Electronic and Associated Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of

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SPECIFICATIONS (Continued)

Military (Continued)

MIL-T-18303	Test Procedures, Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment
MIL-P-18987	Plotting Board, Tactical Display PT-396/AS
MIL-I-19165	Indicator Group AN/APA-125A
MIL-R-21113	Radar Set AN/APS-80(V)
MIL-I-21685	Indicator, Compass Repeater, Rotating Dial, Type MW-2
MIL-I-22075	Indicator, Bearing-Distance-Heading ID-663C/U
MIL-M-22747	Monitor, Error Voltage MX-2230/ASQ and MX-2230A/ASQ
MIL-N-23407	Navigation Set, Radar AN/APN-153(V)
MIL-A-23670	Attitude Heading Reference System AN/ASN-50(*)
MIL-H-27269	Horizontal Situation Indicator Group AF-A24J-1
MIL-M-38510	Microcircuits, General Specification for
MIL-P-55110	Printed Wiring Boards
MIL-D-81358	Display Group, Tactical Data AN/ASA-66
MIL-A-81403	Altimeter-Encoder, AAU-21/A
MIL-N-81497	Navigation Set, Inertial AN/ASN-84
MIL-N-81675	Navigational Set, Omega AN/ARN-99(V)1
MIL-A-81852	Altimeter-Encoder AAU-32/A

STANDARDS

Federal

FED-STD-595	Colors

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STANDARDS (Continued)

Military

MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-463	Definitions and System of Units, Electromagnetic Reference and Electromagnetic Comparability Technology
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipments Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-965	Parts Control Program
MIL-STD-1553	Aircraft, Internal Time Division Command/ Response Multiplex Data Bus
MS17322	Meter, Time Totalizing, Miniature, Digital, 115 Volt, 400 Hertz
MS25212	Control Panel, Console Type, Aircraft Equipment, Basic Dimensions
MS27651	Meter, Time Totalizing Hermetically Sealed, 115V, 400 Hz
MS33638	Case, Instrument, Flange Mounted, Aircraft

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NAVAL AIR SYSTEMS COMMAND

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for
AR-104	Reliability Development Test for Avionics Equipment, Requirements for
AS-l283(AV) AS-1284(AV) AS-1285(AV) AS-1286(AV)	Alphanumerics, Digital Sweep Frequency Generator, and Cass Modifications to the Computer-Recorder Group, Sonar, AN/AQA-7(V) (U) Bearing Bias Correction, Automatic Threshold and Doppler Bearing Tracker Modifications to the Computer-Recorder Group, Sonar AN/AQA-7(V)
EI-632	Avionics Installation Instructions for Navigation-Display Set, AN/ASN-124

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)”

PAGE 7

3.2.1, line 8: Delete “MIL-STD-749” and substitute “MIL-STD-965.”

3.3, line 2: Delete “MIL-P-55640” and substitute “MIL-P-55110.”

PAGE 103

3.5.8.4.6, line 3: Delete “MIL-N-81675(AS)” and substitute “MIL-N-81675.”

PAGE 105

3.5.8.4.9, line 4: Delete “in accordance with MIL-C-81486.”

PAGE 127

4.4.3.1.1, lines 7 and 10: Delete “MIL-STD-781B” and substitute “MIL-STD-781.”

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PAGE 133

6.8, line 2: Delete “MIL-A-81403(AS)” and substitute “MIL-A-81403.”

6.8, line 3: Delete “MIL-A-81852(AS)” and substitute “MIL-A-81852.”

6.8, line 5: Delete “MIL-A-23670A” and substitute “MIL-A-23670.”

6.8, line 9: Delete “MIL-I-22075A” and substitute “MIL-I-22075.”

6.8, line 12: Delete “per MIL-C-18191(WEP).”

6.8, line 18: Delete “per MIL-C-81486.”

6.8, line 19: Delete “per MIL-R-81859.”

6.8, line 25: Delete “MIL-D-81358(AS)” and substitute “MIL-D-81358.”

6.8, line 26: Delete “per MIL-D-85009(AS).”

6.8, line 27: Delete “per MIL-T-19877.”

PAGE 134

Line 2: Delete “MIL-I-21685(WEP)” and substitute “MIL-I-21685.”

PAGE 136

Add an “A” after word APPENDIX in the identifiers at the top of pages 136 through 148.

Change the title to read “APPENDIX A.”

PAGE 148

Add the attached APPENDIX B to form pages 149 through 154.

Preparing Activity
Navy - AS
Project No. 5826-N171

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APPENDIX B

AN/ASN-124 INTERFACE WITH MIL-STD-1553 DATA BUS

10. SCOPE

10.1 Scope. The equipment covered by this appendix shall enable the Navigation Display Set to be interconnected with MIL-STD-1553 type communication busses.

10.2 Classification. The equipment covered by this appendix shall consist of the following items:

Assigned AN Nomenclature	Type Designation	Applicable Paragraph
Controller, Marker Entry	C-9850/ASN-124	3.5.1
Control, Keyer	C-9857/ASN-124	3.5.2
Indicator, Digital Display	ID-2035/ASN-124	3.5.3
Controller, Marker Plotter	C-9855/ASN-124	3.5.4
Converter-Interface	CV-3332/ASN-124	3.5.5
Control, Tactical Display	C-9862/ASN-124	3.5.6
Computer, Navigation	CP-1224A/ASN-124	3.5.7
Converter, Signal Data	CV-3502/ASN-124	30.5.1 Appx. A
Converter, Radar Buffer	CV-3290/ASN-124	3.5.9
Battery Unit	BA-597/ASN-124	3.5.10
Controller, Mode	C-10422/ASN-124	30.5.2 Appx. A
Control-Indicator (2)	C-10421/ASN-124	30.5.3 Appx. A
Interconnecting Box	J-3349/ASN-124	3.5.13

10.3 Associated Equipment. The equipment shall operate to the requirements of this specification with the associated equipment listed in Paragraph 6.8, as modified by paragraphs 30.2 of Appendix A and 40.1.

20. APPLICABLE DOCUMENTS

20.1 General. The following documents form a part of this specification to the extent specified herein. Listed are the issues of the documents in effect for previous procurement. However, in lieu of the issue number listed, the contractor shall use the latest issue in effect of these documents where feasible. If the use of the latest issue will affect design, performance, or interchangeability of any replaceable part, then the issue of the document listed below shall be used.

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APPENDIX B

Standards

MIL-STD-1553	Aircraft Internal Time Division Command/Response Multiplex Data Bus

Commmercial Specifications

PRB-P3B-1001, Rev. 2	P-3B MIL-STD-1553 Multiplex Data Bus Protocol Specification

20.2 Availability of Documents.

20.2.1 Military Documents. When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained from application to the Commanding Officer, Publications and Forms Center, Code 105 Tabor Avenue, Philadelphia, Pennsylvania 19120.

20.2.2 Commercial Specification. When requesting the protocol specification, refer to both title and number. Copies of this protocol specification may be obtained for a fee upon request to PRB Associates, Inc., P.O. Box 66, Great Mills, MD 20634.

30. REQUIREMENTS

The provisions of paragraph 3.0 apply except as modified herein.

30.1 System Definition. The set shall consist of those items of paragraph 10.2 (reference paragraph 3.4.1.1). The system shall operate in conjunction with the associated P-3B aircraft equipment listed in paragraph 10.3. The functional block diagram for this system configuration is shown in Figure 44. The analog output signal distribution diagram for this system configuration is shown in Figure 40 of Appendix A.

30.1.1 Hardware. The CP-1224/ASN-124 shall be converted into a CP-1224A/ASN-124 by replacing the A3 printed circuit card, adding printed circuit cards A14 and A15, replacing the wiring terminal board and providing connectors for the MIL-STD-1553 data bus channels (reference paragraph 3.4.1.1.1).

30.1.2 Software. The system shall be programmed to perform the following basic functions (reference paragraph 3.4.1.1.2):

(1) Navigation
(2) Control Panel Interface
(3) Display Interface
(4) System Self Test and Fault Isolation
(5) Control of MIL-STD-1553 Data Bus

30.1.3 Electrical Connections. All connections of paragraph 3.5.7.7, as modified by 30.5.2.8 and 30.5.3.8 of Appendix A, shall be provided with the following additions:

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REF DES	REC TYPE
J11	PN 165675-01
J12	PN 165675-01

30.1.4 Memory Protection. A memory protect capability shall be provided in the logic of the memory control circuits and shall operate as follows (reference paragraph 3.5.7.4.3.1):

The computer, using an I/O channel, shall under software control load a protect boundary into a dedicated memory protect register. This boundary shall define a memory location below which no write operation is allowed. If a write is attempted into an address below the protect boundary, the memory cycle shall be aborted and an interrupt shall be generated.

30.1.5 Input/Output. The NC shall contain the I/O specified in 3.5.7.4.4. In addition, the NC shall contain an interface that conforms to MIL-STD-1553. The interface shall allow the NC to function as either a Bus Controller, or a Remote Terminal. This I/O channel shall function as a dual redundant unit as identified by MIL-STD-1553.

30.1.6 Bootstrap Load. The NC shall have a bootstrap capability. The bootstrap shall be able to load an operational program into core memory from the MIL-STD-1553 Data Bus. The bootstrap shall operate regardless of whether any software is currently loaded into memory, and without the need to have external test equipment connected.

40. NOTES

40.1 Associated Equipment

Teletypewriter Set	AN/AGC-9(V)1
Recorder/Reproducer	RD-461/A

☆ U.S. GOVERNMENT PRINTING OFFICE: 1982-505-022/2165

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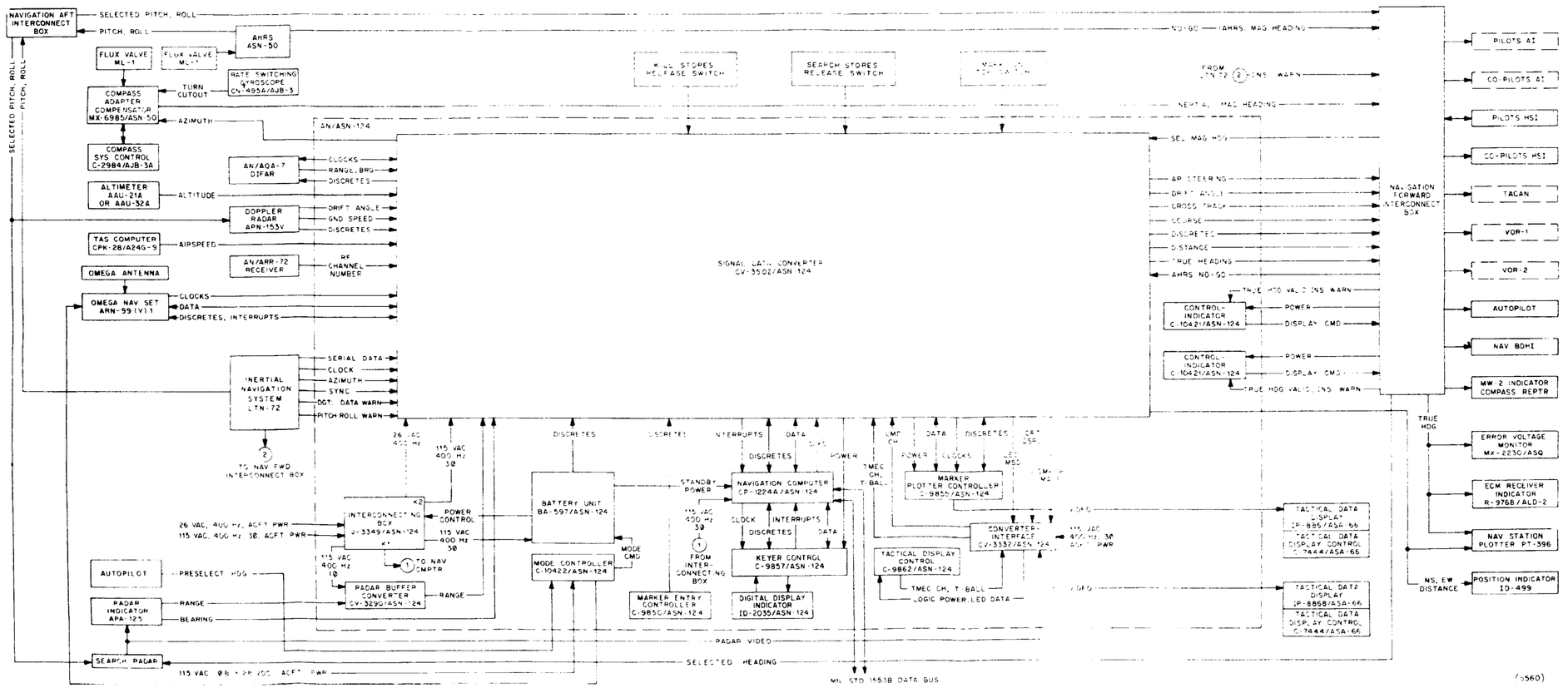


Figure 44. Functional Block Diagram,
Navigation-Display Set, AN/ASN-124, ARINC-561 INS Interconnect, MIL-STD-1553 Data Bus

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